Breaking MT7621 Limits - 300Mbps+ Stability via Driver-Level Hard Binding and In-place NAPI Polling

Software Stack & Methodology

- **Hardware**: MediaTek MT7621AT (**OC 1.0GHz**) + MT7915E (WiFi 6).
- **Kernel/Driver**: **Linux Kernel 5.4** + **OpenWrt/mt76 (2022 Version)**.
- **Critical Patches**: Latest **Memory Leak Patches**, **WiFi 5 AMSDU/MPDU Tuning**, and **Airtime Stat Correction**.

300M Packet Milestone Performance Scorecard (SU Total: 301,148,189)

Performance Metric	Detailed Description	Real-time Value	Status Determination
DL Throughput	iperf3 -R Downlink Average	301.83 ~ 285.63 Mbps	Ultra-Stable
SU Total	Total Successful Single-User MPDUs	301,148,189	Milestone Reached
Squeeze Δ	NAPI Budget Exhaustion Delta	0	Absolute CPU Surplus
ksoftirqd Load	Software Interrupt Process Load	0%	No Softirq Storm
kworker Load	Async Workqueue Process Load	None	No Token Backlog
MSDU-1 Ratio	Single-packet (Non-aggregated) Ratio	1.4%	Extreme Timing Precision
5G Temperature	MT7915 PHY Real-time Temp	63.0 ℃	Thermal Equilibrium

Free Memory	**Remaining Physical RAM**	**62.73 MB**	**Mem-Steady (No Sludge)**

Technical Deep Dive

A. Why Pinning mt76-tx to CPU 2 is Mandatory?

- **Role of mt76-tx**: This thread handles **AMSDU aggregation** and **DMA injections**. It relies on **HRTIMER** microsecond precision. If the CPU running `mt76-tx` is also involved in packet parsing, high-priority **softirqd** tasks will preempt the CPU, shattering the **Aggregation Window**.
- **Consequence**: Interrupted aggregation leads to forced **MSDU-1** flushes, skyrocketing interrupt frequency, and eventual system collapse.

B. How RPS=0 (NAPI In-place Polling) Eliminates kworker Loops?

- **NAPI Polling Nature**: **RPS=0** utilizes **In-place NAPI Processing**. CPU 3 handles packets directly within the current interrupt context, entering a `Poll` loop to "harvest" locally.
- **Breaking the Deadlock**: This eliminates **IPI synchronization** overhead. Instant **Token** reclamation within a single-core closed loop causes `kworker` to vanish (**None**) due to zero trigger conditions.

C. Why SLAB is Preferred over SLUB?

- **Cache Coloring**: On the cache-constrained MT7621, **SLAB** provides superior cache coloring, significantly reducing **Cache Aliasing** during high-frequency `skbuff` allocations.
- **Fragmentation Control**: While **SLUB** scales well on large systems, it causes aggressive **external fragmentation** on **256MB** MIPS devices under 300Mbps+ stress.
- **Determinism**: **SLAB's** per-CPU local queues ensure that `skbuff` allocation/deallocation remains localized to CPU 3, avoiding the global page-reclamation latencies of SLUB and maintaining the **Squeeze=0** threshold.

Appendix: Detailed data (Please disregard runtime statistics - this is based on multiple monitoring script modifications and restarts, and is subject to real-time statistical errors; download speed is +/- 20MB/s)

```
==========================================================================================
  MT7915 深度监控 v9.5 (流式对齐修正版)    目标设备: 9c:b6:d0:c2:85:07
==========================================================================================
 CPU 占用: 58.5%  64.6%  47.4%  86.8%
 系统负载: 4.05 3.89 3.99     | 运行时间: 02:01:43
------------------------------------------------------------------------------------------
 物理节点: phy1 (wlan1-1)  信号强度: -53 [-54, -60] dBm
 5G 温度: 63.0℃ (限值:95.0℃) | 空闲内存: 60.30 MB
------------------------------------------------------------------------------------------
 [SLAB 关键内存池 (Active/Total)]
  skbuff_fclone_cache: 295/470
  skbuff_head_cache : 1218/2058
------------------------------------------------------------------------------------------
 [每秒吞吐量 (Mbps)]  下载(DL): 286.72 Mbps | 上传(UL): 1.49  Mbps
------------------------------------------------------------------------------------------
 [物理层状态]
  协商速率: 650.0 MBit/s VHT-MCS 7 80MHz short GI VHT-NSS
  队列深度: 2          | BA MISS Δ: 474
  累计 BA MISS: 614066     (8.28%) | 采样周期: 4.4100s
------------------------------------------------------------------------------------------
  [MSDU 聚合(SU 基准)] SU Total: 322975509
  M1:1.4% M2:47.2% M3:11.7% M4:0.0% | 1-4合计: 194742623 (60.30%)
------------------------------------------------------------------------------------------
  RPS 状态: wlan1-1:[0] wlan1:[0] wlan0:[0] br-lan_dhcp:[0] br-lan:[-] lo:[0]
------------------------------------------------------------------------------------------
 [ksoftirqd 占用]: CPU0: 0.0%  CPU3: 0.0%  CPU1: 0.0%  CPU2: 0.0%
 [kworker 占用]: None
------------------------------------------------------------------------------------------
CPU   | Packets (Δ)          | Dropped (Δ)          | Squeeze (Δ)          | IRQs (Δ)
CPU0  | 94817372   (+2963 ) | 0          (+0    ) | 421        (+0    ) | 287752     (+0    )
CPU1  | 90072255   (+2979 ) | 0          (+0    ) | 232        (+0    ) | 184727     (+0    )
CPU2  | 9034341    (+235  ) | 0          (+0    ) | 0          (+0    ) | 3419       (+0    )
CPU3  | 23697868   (+381  ) | 0          (+0    ) | 0          (+0    ) | 206888     (+0    )
------------------------------------------------------------------------------------------
IRQ   | CPU0(Δ Tot)       | CPU1(Δ Tot)       | CPU2(Δ Tot)       | CPU3(Δ Tot)       | NAME
------------------------------------------------------------------------------------------
8     | 100    2999683    | 100    2999667    | 100    2999659    | 100    2999651    | MIPS GIC Local 1 tim
9     | 243    5587571    |                   |                   |                   | MIPS GIC 63 IPI call
10    |                   | 24     1096352    |                   |                   | MIPS GIC 64 IPI call
11    |                   |                   | 58     7364426    |                   | MIPS GIC 65 IPI call
12    |                   |                   |                   | 19     1138760    | MIPS GIC 66 IPI call
13    | 474    12964501   |                   |                   |                   | MIPS GIC 67 IPI resc
14    |                   | 489    13053347   |                   |                   | MIPS GIC 68 IPI resc
15    |                   |                   | 1587   46435390   |                   | MIPS GIC 69 IPI resc
16    |                   |                   |                   | 79     3576733    | MIPS GIC 70 IPI resc
22    |        43844      |                   |                   |                   | MIPS GIC 10 1e100000
24    |        105        |                   | 116    4657340    |                   | MIPS GIC 11 mt7915e-
25    |        1044       |                   |                   | 623    17431415   | MIPS GIC 31 mt7915e
ERR   | 1      38974      |                   |                   |                   |
```